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A03 Case Study

The Liverpool Smart Pedestrian project was a smart visual sensing system developed as part of a pilot deployment in Liverpool, Australia. The goal of the project was to apply computer vision and deep neural networks to create and evaluate an edge-computing device capable of tracking real-time multi-modal transportation activity while maintaining citizen privacy. The sensor was tested using a town center dataset and supported by the Agnosticity framework, which enabled the collection, storage, and access of data from multiple heterogeneous sensors. Overall, the project sought to support Liverpool's rapid population growth while addressing broader social, sustainability, and traffic management challenges.

This initiative was designed to provide an innovative and non-intrusive solution for urban data collection in New South Wales. Liverpool was experiencing significant expansion in housing, commercial development, and educational infrastructure, with projections suggesting an increase of roughly 30,000 additional pedestrians per day. This rapid growth made the city an ideal environment for experimenting with technologies aimed at monitoring and improving future traffic conditions.

Two community workshops conducted at the beginning of the project helped define the system requirements. Feedback from urban planners highlighted the need for a sensor capable of multi-modal detection and tracking of pedestrians, cyclists, and vehicles. Privacy compliance was a central concern, as the system needed to ensure that personal data could not be compromised. Additionally, the planners emphasized the importance of leveraging existing city infrastructure and supporting the integration of new sensors regardless of the technology used. This requirement ensured long-term scalability and interoperability. Key constraints included keeping deployment costs low while ensuring that the solution remained reproducible and easy to expand.

The authors selected the NVIDIA Jetson TX2 as the primary embedded computing platform due to its high performance and power efficiency. The TX2 provided specialized hardware for accelerating neural network computations and supported image processing workloads on Ubuntu 16.04 LTS. Communication was handled through a Pycom LoPy 4 module, which enabled LoRaWAN transmission on the AS923 frequency plan used in Australia while also supporting other LoRaWAN frequency plans. To display the sensor's operational state, the system included a DuinoTECH XC-4384 compact monochrome OLED module. All components were enclosed in a waterproof IP67-rated aluminum heat sink case designed for outdoor deployment and thermal dissipation.

For object detection, the project employed YOLOv3, which at the time represented a strong balance between speed and accuracy for real-time traffic flow monitoring. Since the system needed to operate on an embedded edge device, selecting an algorithm capable of efficient inference was essential. YOLOv3, a fully convolutional deep neural network, was able to detect objects at three different scales and thus met the project's requirements for identifying pedestrians, vehicles, and cyclists. Its implementation in PyTorch allowed it to take advantage of the Jetson TX2's CUDA cores. Additionally, YOLO used Non-Maximal Suppression to remove duplicate detections, improving efficiency by reducing redundant computation.

The edge-computing paradigm was critical to the system's privacy goals. By processing video locally rather than transmitting full image feeds over networks, the sensor ensured that raw CCTV footage remained within the local infrastructure. Only denatured, privacy-preserving data outputs were produced, reducing both bandwidth demands and privacy risks.

System validation was conducted using the first 4,500 frames (approximately three minutes) of the evaluation video. In this segment, 230 pedestrians were manually annotated to establish ground truth. When compared to these annotations, the model achieved a mean accuracy of 69% and a median relative error of 33%. One major issue was the sensor's tendency to underestimate detections, largely due to occlusions common in crowded environments. Overlapping bounding boxes also posed challenges for YOLO's Non-Maximal Suppression, reducing detection performance.

Performance analysis showed that frame rate was primarily influenced by the number of detected objects: more detections led to lower FPS due to the computational burden of the SORT tracking algorithm. SORT did not fully utilize the Jetson TX2's GPU resources, resulting in frequent underutilization, with GPU usage dropping from 100% to below 40%. Despite this, system temperature remained stable during operation.

The project included both indoor and outdoor deployment experiments. The indoor deployment occurred during an emergency evacuation scenario and successfully detected movement into and out of the building. Importantly, the system did not falsely detect individuals remaining inside during the alarm event, indicating that the building was effectively evacuated. A secondary observation peak revealed that fewer people re-entered the building even after firefighters deemed it safe, suggesting the sensor could detect abnormal crowd behavior. Such capability could enhance monitoring in sensitive environments like schools or government facilities.

The outdoor deployment in Liverpool also proved effective. The sensor successfully detected pedestrians, vehicles, and even the small number of cyclists traveling through the area. This type of monitoring could help inform decisions about resource deployment, such as placing crossing guards, police presence, or sanitation crews at optimal times. Additionally, the collected data could support urban planning decisions, such as identifying high-traffic areas suitable for new shops or services.

For future work, the authors emphasized improving detection and tracking performance. They suggested that quantifying detection error could help address duplicate detections and overall accuracy limitations. They also noted that upgrading to NVIDIA's Xavier platform could provide significant performance improvements over the Jetson TX2.

Since the project's publication in 2019, several technological advancements could further enhance systems like this one. Modern hardware such as the Jetson Orin (released in 2022) offers 10–20× faster inference performance and supports higher camera throughput. Dedicated accelerators like Google Coral TPUs, Intel Movidius chips, or FPGA-based solutions could also reduce power consumption and latency. Detection algorithms have also improved substantially, with newer YOLO versions offering better small-object recognition, improved crowd detection, and stronger robustness under occlusions and poor lighting. Tracking could likewise benefit from modern approaches such as DeepSORT, ByteTrack, OC-SORT, or BoT-SORT, which provide more stable identity tracking and fewer false negatives. Finally, newer sensor modalities like event-based cameras or low-cost LiDAR could further improve performance, especially in challenging weather or nighttime conditions.

Overall, the Liverpool Smart Pedestrian project was highly successful for its time. The system demonstrated strong real-world performance in both indoor safety monitoring and outdoor traffic observation, despite limitations related to occlusion and duplicate detections. Innovations like this have the potential to significantly improve city safety initiatives and urban planning efforts. With modern upgrades in hardware, detection models, and tracking methods, projects of this kind could become even more accurate, scalable, and impactful today.

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